

CBSE Class 12 Maths — Half-Yearly Predicted Paper (Set 6 of 10)

This is Set 6 of the Half-Yearly Predicted Paper series, formulated strictly according to the latest CBSE competency-based blueprint for the 2026-27 academic session. This paper strictly tests the ~80% half-yearly syllabus, intentionally excluding Linear Programming and Probability. Every question is uniquely selected from high-weightage topics and frequently repeated PYQ trends.

SECTION A: Objective Type Questions (1 Mark Each x 20 = 20 Marks)

- Q1.** Let $A = \{a, b, c\}$. What is the minimum number of ordered pairs that must be added to the relation $R = \{(a, a), (b, b), (a, b)\}$ to make it an equivalence relation? * **Chapter:** Relations and Functions * **Confidence: High**
- Q2.** Evaluate the exact value of $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: Very High**
- Q3.** If A is a non-singular square matrix of order 3 such that $|A| = 8$, find the value of $|A \cdot \text{adj}A|$. * **Chapter:** Matrices & Determinants * **Confidence: Very High**
- Q4.** Given that the determinant equation $\begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix} = \begin{vmatrix} 4 & -2 \\ 3 & 1 \end{vmatrix}$ holds true, find the positive value of x . * **Chapter:** Determinants * **Confidence: High**
- Q5.** What is the derivative of $\log_a x$ with respect to x (where $a > 0, a \neq 1$)? * **Chapter:** Continuity and Differentiability * **Confidence: Medium**
- Q6.** Determine the rate of change of the perimeter of a circle with respect to its radius r . * **Chapter:** Application of Derivatives * **Confidence: High**
- Q7.** Evaluate the indefinite integral: $\int \frac{2^x}{\sqrt{1-4^x}} dx$. * **Chapter:** Integrals * **Confidence: Very High**
- Q8.** Evaluate the definite integral: $\int_0^{\pi/2} \sin 2x dx$. * **Chapter:** Integrals * **Confidence: High**
- Q9.** State the order and degree (if defined) of the differential equation: $y''' + 2(y'')^2 + y' = \log(y'')$. * **Chapter:** Differential Equations * **Confidence: Very High**
- Q10.** Write the general solution of the differential equation $\frac{dy}{dx} = \frac{1}{x^2+1}$. * **Chapter:** Differential Equations * **Confidence: Medium**
- Q11.** Find the value of p for which the vectors $\vec{a} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\vec{b} = \hat{i} + p\hat{j} + 3\hat{k}$ are strictly parallel. * **Chapter:** Vector Algebra * **Confidence: Very High**
- Q12.** What is the magnitude squared, $|\hat{i} - \hat{j}|^2$, of the given vector? * **Chapter:** Vector Algebra * **Confidence: High**
- Q13.** Without calculation, state the shortest distance between the two lines in space given by $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{a} + \mu\vec{c}$. * **Chapter:** Three-Dimensional Geometry * **Confidence: High**
- Q14.** Write the exact intercepts made by the plane $2x - 3y + 4z = 12$ on the x, y , and z coordinate axes respectively. * **Chapter:** Three-Dimensional Geometry * **Confidence: Very High**
- Q15.** Let $f(x) = \frac{x-1}{x+1}$, where $x \neq -1$. Find the simplified expression for the composition $f(f(x))$. * **Chapter:** Relations and Functions * **Confidence: High**
- Q16.** For any square matrix A with real number entries, what specific type of matrix is $A + A^T$? * **Chapter:** Matrices * **Confidence: Very High**
- Q17.** Find the slope of the tangent to the curve $y = x^3 - x$ at the point where $x = 2$. * **Chapter:** Application of Derivatives * **Confidence: High**
- Q18.** Evaluate the indefinite integral: $\int e^x \sec x(1 + \tan x) dx$. * **Chapter:** Integrals * **Confidence: Very High**
- Q19. (Assertion-Reason): Assertion (A):** The function $f(x) = \sin |x|$ is continuous for all real values of x . **Reason (R):** Both the modulus function and the sine function are continuous everywhere, and the composition of two continuous functions is a continuous function. Choose the correct option. * **Chapter:** Continuity and Differentiability * **Confidence: Very High**
- Q20. (Assertion-Reason): Assertion (A):** For any two vectors $\vec{a}$ and $\vec{b}$, the inequality $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$ always holds true. **Reason (R):** The sum of the lengths of any two sides of a triangle is greater than or equal to the length of the third side. Choose the correct option. * **Chapter:** Vector Algebra * **Confidence: High**

SECTION B: Very Short Answer Type Questions (2 Marks Each x 5 = 10 Marks)

- Q21.** Prove algebraically that $2 \tan^{-1}\left(\frac{1}{3}\right) = \tan^{-1}\left(\frac{3}{4}\right)$. * **Chapter:** Inverse Trigonometric Functions * **Confidence: High**
- Q22.** Find the values of x and y if $2 \begin{pmatrix} x & 5 \\ 7 & y-3 \end{pmatrix} + \begin{pmatrix} 3 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 7 & 6 \\ 15 & 14 \end{pmatrix}$. * **Chapter:** Matrices * **Confidence: Very High**

Q23. Solve the differential equation by variable separation: $\log_e \left(\frac{dy}{dx} \right) = ax + by$. * **Chapter:** Differential Equations *

Confidence: High

Q24. The volume of a cube is increasing at the constant rate of $9 \text{ cm}^3/\text{s}$. How fast is its surface area increasing when the length of an edge is 10 cm ? * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q25. Find the area of the triangle whose adjacent sides are determined by the vectors $\vec{a} = 3\hat{i} + 4\hat{j}$ and $\vec{b} = -3\hat{i} + 7\hat{j}$. * **Chapter:** Vector Algebra * **Confidence: Medium**

SECTION C: Short Answer Type Questions (3 Marks Each x 6 = 18 Marks)

Q26. Check whether the relation R in the set of real numbers $\mathbb{R}$ defined by $R = \{(a, b) : a \leq b^3\}$ is reflexive, symmetric, or transitive. Justify your answer with counterexamples where necessary. * **Chapter:** Relations and Functions * **Confidence: Very High**

Q27. Find the inverse of the matrix $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and use it to solve the system of linear equations: $2x - y = 5$ and $3x + 4y = 2$. * **Chapter:** Determinants * **Confidence: High**

Q28. Differentiate the function $y = (\sin x)^x + \sin^{-1}(\sqrt{x})$ with respect to x . * **Chapter:** Continuity and Differentiability * **Confidence: Very High**

Q29. Evaluate the indefinite integral using the method of partial fractions: $\int \frac{x^2}{(x^2+1)(x^2+4)} dx$. * **Chapter:** Integrals * **Confidence: Very High**

Q30. Find the particular solution of the differential equation $(1 + e^{2x})dy + (1 + y^2)e^x dx = 0$, given that $y = 1$ when $x = 0$. * **Chapter:** Differential Equations * **Confidence: High**

Q31. Find the exact shortest distance between the two skew lines defined by $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$ and $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$. * **Chapter:** Three-Dimensional Geometry * **Confidence: Very High**

SECTION D: Long Answer Type Questions (5 Marks Each x 4 = 20 Marks)

Q32. Find the matrix product AB , where $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{pmatrix}$. Use this product to solve the following system of linear equations: $x - y + z = 4$, $x - 2y - 2z = 9$, $2x + y + 3z = 1$. * **Chapter:** Matrices and Determinants * **Confidence: Very High**

Q33. Prove mathematically using calculus that the volume of the largest cone that can be inscribed in a sphere of given radius R is exactly $\frac{8}{27}$ of the volume of the sphere. * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q34. Evaluate the following definite integral using limit properties: $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$. * **Chapter:** Integrals * **Confidence: High**

Q35. Using the method of integration, sketch the region and find the exact area bounded by the triangle whose vertices are $(-1, 1)$, $(0, 5)$, and $(3, 2)$. * **Chapter:** Application of Integrals * **Confidence: Very High**

SECTION E: Case-Study Based Questions (4 Marks Each x 3 = 12 Marks)

Q36. Case Study 1: Matrices (Factory Production) Two factories, Factory A and Factory B, produce three types of sports shoes: sneakers, cleats, and runners. The daily production of Factory A consists of 20, 30, and 40 units of sneakers, cleats, and runners respectively. Factory B produces 30, 20, and 50 units of the same respectively per day. The market selling price per pair is ₹1000 for sneakers, ₹1500 for cleats, and ₹1200 for runners. (i) Construct a 2×3 matrix representing the daily production of the two factories. (1 Mark) (ii) Construct a column matrix representing the selling prices of the shoes. (1 Mark) (iii) Using matrix multiplication, calculate the total daily revenue generated by Factory A and Factory B individually. (2 Marks) * **Chapter:** Matrices * **Confidence: High**

Q37. Case Study 2: Application of Derivatives (Manufacturing Cost) A company manufactures x number of identical water tanks per month. The monthly fixed cost is ₹10,000, and the variable cost per tank is ₹(500 + x). The selling price per tank in the market is fixed at ₹2000. (i) Write the mathematical Profit function $P(x)$ in terms of the number of tanks x . (1 Mark) (ii) Find the marginal profit function $P'(x)$. (1 Mark) (iii) Use calculus to find the exact number of tanks that must be produced and sold to maximize the monthly profit, and calculate this maximum profit. (2 Marks) * **Chapter:** Application of Derivatives * **Confidence: Very High**

Q38. Case Study 3: 3D Geometry (Aviation Flight Paths) Two airplanes follow linear flight paths in a 3D simulated airspace. Air Traffic Control maps Airplane 1's path using the Cartesian equation $\frac{x-1}{2} = \frac{y-2}{-1} = \frac{z-3}{2}$. Airplane 2 follows the path given by $\frac{x-2}{3} = \frac{y-4}{2} = \frac{z-5}{6}$. (i) Find the direction cosines of the flight path of Airplane 1. (1 Mark) (ii) Calculate the exact angle between the two flight paths. (1 Mark) (iii) Check mathematically whether the two flight paths intersect each other, or if they are skew lines. Show your working. (2 Marks) * **Chapter:** Three-Dimensional Geometry * **Confidence: Very High**

Answer Key

Q1. Two pairs: (c, c) to make it reflexive, and (b, a) to make it symmetric. **Q2.**

$\sin\left(\frac{\pi}{3} - \left(-\frac{\pi}{6}\right)\right) = \sin\left(\frac{\pi}{3} + \frac{\pi}{6}\right) = \sin\left(\frac{\pi}{2}\right) = 1$. **Q3.** We know $A \cdot \text{adj}A = |A|I$. Therefore

$|A \cdot \text{adj}A| = ||A|I| = |A|^n = 8^3 = 512$. **Q4.**

$2x^2 - 24 = 4 - (-6) \Rightarrow 2x^2 - 24 = 10 \Rightarrow 2x^2 = 34 \Rightarrow x^2 = 17 \Rightarrow x = \sqrt{17}$ (taking positive value). **Q5.**

$\frac{d}{dx}(\log_a x) = \frac{d}{dx}\left(\frac{\log_e x}{\log_e a}\right) = \frac{1}{x \log_e a}$. **Q6.** Perimeter $P = 2\pi r \Rightarrow \frac{dP}{dr} = 2\pi$. **Q7.** Let $2^x = t \Rightarrow 2^x \ln 2 dx = dt$.

Integral becomes $\frac{1}{\ln 2} \int \frac{dt}{\sqrt{1-t^2}} = \frac{1}{\ln 2} \sin^{-1}(t) + C = \frac{1}{\ln 2} \sin^{-1}(2^x) + C$. **Q8.**

$\left[-\frac{\cos 2x}{2}\right]_0^{\pi/2} = -\frac{1}{2}(\cos \pi - \cos 0) = -\frac{1}{2}(-1 - 1) = 1$. **Q9.** Order is 3 (highest derivative is y'''). Degree is not defined because the equation is not a polynomial in its derivatives (due to the $\log(y'')$ term). **Q10.**

$\int dy = \int \frac{1}{x^2+1} dx \Rightarrow y = \tan^{-1} x + C$. **Q11.** For parallel vectors, direction ratios are proportional: $\frac{3}{1} = \frac{2}{p} = \frac{9}{3}$. Thus,

$3 = \frac{2}{p} \Rightarrow p = \frac{2}{3}$. **Q12.** $|\hat{i} - \hat{j}|^2 = (\sqrt{1^2 + (-1)^2})^2 = (\sqrt{2})^2 = 2$. **Q13.** 0. (Both lines pass through the point with position vector $\vec{a}$, meaning they intersect at that point). **Q14.** Divide by 12: $\frac{x}{6} + \frac{y}{-4} + \frac{z}{3} = 1$. Intercepts are 6, -4, 3. **Q15.**

$f(f(x)) = \frac{\frac{x-1}{x+1}-1}{\frac{x-1}{x+1}+1} = \frac{(x-1)-(x+1)}{(x-1)+(x+1)} = \frac{-2}{2x} = -\frac{1}{x}$. **Q16.** Symmetric Matrix. (Since

$(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$. **Q17.** $y' = 3x^2 - 1$. At $x = 2$, slope = $3(4) - 1 = 11$. **Q18.**

$\int e^x (\sec x + \sec x \tan x) dx$. This is of the form $\int e^x [f(x) + f'(x)] dx$ where $f(x) = \sec x$. Answer is $e^x \sec x + C$. **Q19.**

Option A. (Both A and R are true, and R is the correct explanation of A). **Q20.** Option A. (Both A and R are true, and R is the correct explanation of A - this represents the Triangle Inequality for vectors).

Q21. Let LHS = $\tan^{-1}\left(\frac{2(1/3)}{1-(1/3)^2}\right) = \tan^{-1}\left(\frac{2/3}{8/9}\right) = \tan^{-1}\left(\frac{2}{3} \times \frac{9}{8}\right) = \tan^{-1}\left(\frac{3}{4}\right) = \text{RHS}$. **Q22.** Comparing components:

$2x + 3 = 7 \Rightarrow 2x = 4 \Rightarrow x = 2$. And $2(y - 3) + 2 = 14 \Rightarrow 2y - 6 + 2 = 14 \Rightarrow 2y = 18 \Rightarrow y = 9$. **Q23.**

$\frac{dy}{dx} = e^{ax+by} = e^{ax} \cdot e^{by}$. Separating variables: $e^{-by} dy = e^{ax} dx$. Integrating: $\frac{e^{-by}}{-b} = \frac{e^{ax}}{a} + C$. **Q24.** Volume

$V = x^3 \Rightarrow \frac{dV}{dt} = 3x^2 \frac{dx}{dt} = 9 \Rightarrow \frac{dx}{dt} = \frac{3}{x^2}$. Surface Area $S = 6x^2 \Rightarrow \frac{dS}{dt} = 12x \frac{dx}{dt} = 12x \left(\frac{3}{x^2}\right) = \frac{36}{x}$. At

$x = 10$, rate is $3.6 \text{ cm}^2/\text{s}$. **Q25.** Area = $\frac{1}{2} |\vec{a} \times \vec{b}|$. $\vec{a} \times \vec{b} = (3\hat{i} + 4\hat{j}) \times (-3\hat{i} + 7\hat{j}) = 21\hat{k} - (-12\hat{k}) = 33\hat{k}$. Magnitude is 33. Area = $\frac{33}{2} = 16.5$ sq units.

Q26. Not Reflexive: Let $a = 1/2$. $1/2 \not\leq (1/2)^3 = 1/8$. **Not Symmetric:** $1 \leq 2^3$, but $2 \not\leq 1^3$. **Not Transitive:** Let

$a = 2, b = 1.5, c = 1.2$. $2 \leq (1.5)^3 = 3.375$. $1.5 \leq (1.2)^3 = 1.728$. But $2 \not\leq (1.2)^3 = 1.728$. **Q27.** $|A| = 8 - (-3) = 11$.

$\text{adj}A = \begin{pmatrix} 4 & 1 \\ -3 & 2 \end{pmatrix}$. $A^{-1} = \frac{1}{11} \begin{pmatrix} 4 & 1 \\ -3 & 2 \end{pmatrix}$. For equations,

$AX = B \Rightarrow X = A^{-1}B = \frac{1}{11} \begin{pmatrix} 4 & 1 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 22 \\ -11 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. $x = 2, y = -1$. **Q28.** Let

$u = (\sin x)^x \Rightarrow \log u = x \log(\sin x) \Rightarrow \frac{1}{u} \frac{du}{dx} = 1 \cdot \log(\sin x) + x \cot x \Rightarrow \frac{du}{dx} = (\sin x)^x (\log \sin x + x \cot x)$.

Let $v = \sin^{-1}(\sqrt{x}) \Rightarrow \frac{dv}{dx} = \frac{1}{\sqrt{1-(\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2\sqrt{x-x^2}}$. Finally, $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$. **Q29.** Substitute $x^2 = t$ (only for

partial fractions). $\frac{t}{(t+1)(t+4)} = \frac{-1/3}{t+1} + \frac{4/3}{t+4}$. Back substitute $t = x^2$:

$\int \left(-\frac{1}{3(x^2+1)} + \frac{4}{3(x^2+4)}\right) dx = -\frac{1}{3} \tan^{-1} x + \frac{2}{3} \tan^{-1}\left(\frac{x}{2}\right) + C$. **Q30.** $\frac{dy}{1+y^2} = -\frac{e^x}{1+e^{2x}} dx$. Integrating: $\tan^{-1} y = -\int \frac{dt}{1+t^2}$

(where $t = e^x$) $\Rightarrow \tan^{-1} y = -\tan^{-1}(e^x) + C$. At

$x = 0, y = 1 \Rightarrow \tan^{-1}(1) = -\tan^{-1}(1) + C \Rightarrow \frac{\pi}{4} = -\frac{\pi}{4} + C \Rightarrow C = \frac{\pi}{2}$. Sol: $\tan^{-1} y + \tan^{-1}(e^x) = \frac{\pi}{2}$.

Q31. $\vec{a}_2 - \vec{a}_1 = -4\hat{i} - 6\hat{j} - 8\hat{k}$. $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 7 & -6 & 1 \end{vmatrix} = 4\hat{i} + 6\hat{j} + 8\hat{k}$.

$SD = \frac{|(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|} = \frac{|-16 - 36 - 64|}{\sqrt{16+36+64}} = \frac{116}{\sqrt{116}} = \sqrt{116} = 2\sqrt{29}$ units.

Q32. $AB = \begin{pmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{pmatrix} = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} = 8I$. Thus, $A^{-1} = \frac{1}{8}B$. Given system is $AX = \begin{pmatrix} 4 \\ 9 \\ 1 \end{pmatrix}$.

$X = A^{-1}C = \frac{1}{8}B \begin{pmatrix} 4 \\ 9 \\ 1 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} -16 + 36 + 4 \\ -28 + 9 + 3 \\ 20 - 27 - 1 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 24 \\ -16 \\ -8 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -1 \end{pmatrix}$. $x = 3, y = -2, z = -1$. **Q33.** Let radius of

cone base be r , height be h , and x be the distance from the sphere center to the cone base. Then $h = R + x$ and

$r^2 = R^2 - x^2$. Volume $V = \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi (R^2 - x^2)(R + x)$.

$\frac{dV}{dx} = \frac{\pi}{3} [(-2x)(R + x) + (R^2 - x^2)(1)] = \frac{\pi}{3} (R + x)(-2x + R - x) = \frac{\pi}{3} (R + x)(R - 3x)$. Setting to zero gives

$x = R/3$. Then $h = \frac{4R}{3}$ and $r^2 = \frac{8R^2}{9}$. $V_{\max} = \frac{1}{3} \pi \left(\frac{8R^2}{9}\right) \left(\frac{4R}{3}\right) = \frac{32\pi R^3}{81} = \frac{8}{27} \left(\frac{4}{3} \pi R^3\right) = \frac{8}{27} V_{\text{sphere}}$. $V''(R/3) < 0$

proves maximum. **Q34.** Let $I = \int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$. Use property $\int_0^a f(x) = \int_0^a f(a-x)$. Adding I to itself yields

$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{\sin x \cos x}{\sin^4 x + \cos^4 x} dx$. Divide num & den by $\cos^4 x$ to get $2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{\tan x \sec^2 x}{\tan^4 x + 1} dx$. Let

$t = \tan^2 x \Rightarrow dt = 2 \tan x \sec^2 x dx$. Limits change to $0 \rightarrow \infty$. $2I = \frac{\pi}{4} \int_0^{\infty} \frac{dt}{t^2 + 1} = \frac{\pi}{4} [\tan^{-1} t]_0^{\infty} = \frac{\pi}{4} \left(\frac{\pi}{2}\right) = \frac{\pi^2}{8}$.

Thus, $I = \frac{\pi^2}{16}$. **Q35.** Vertices: $A(-1, 1), B(0, 5), C(3, 2)$. Equations of sides: $AB \Rightarrow y = 4x + 5$; $BC \Rightarrow y = -x + 5$; $AC \Rightarrow y = \frac{x+5}{4}$. Area

$$\begin{aligned}
 &= \int_{-1}^0 (y_{AB} - y_{AC}) dx + \int_0^3 (y_{BC} - y_{AC}) dx = \int_{-1}^0 \left(4x + 5 - \frac{x+5}{4}\right) dx + \int_0^3 \left(-x + 5 - \frac{x+5}{4}\right) dx \\
 &= \int_{-1}^0 \left(\frac{15x+15}{4}\right) dx + \int_0^3 \left(\frac{-5x+15}{4}\right) dx = \frac{15}{4} \left[\frac{x^2}{2} + x\right]_{-1}^0 + \frac{5}{4} \left[\frac{-x^2}{2} + 3x\right]_0^3 = \frac{15}{4} \left[0 - \left(\frac{1}{2} - 1\right)\right] + \frac{5}{4} \left[\frac{-9}{2} + 9 - 0\right] = \frac{15}{8} + \frac{45}{8} = \\
 &\text{sq units.}
 \end{aligned}$$

Q36. (i) Production matrix $P = \begin{pmatrix} 20 & 30 & 40 \\ 30 & 20 & 50 \end{pmatrix}$. (ii) Price matrix $S = \begin{pmatrix} 1000 \\ 1500 \\ 1200 \end{pmatrix}$. (iii) Total Revenue

$$R = P \times S = \begin{pmatrix} 20(1000) + 30(1500) + 40(1200) \\ 30(1000) + 20(1500) + 50(1200) \end{pmatrix} = \begin{pmatrix} 20000 + 45000 + 48000 \\ 30000 + 30000 + 60000 \end{pmatrix} = \begin{pmatrix} 113000 \\ 120000 \end{pmatrix}.$$

Factory A generates ₹1,13,000 and Factory B generates ₹1,20,000 per day. **Q37.** (i) Cost $C(x) = 10000 + x(500 + x) = 10000 + 500x + x^2$.

Revenue $R(x) = 2000x$. Profit $P(x) = R(x) - C(x) = 2000x - (10000 + 500x + x^2) = 1500x - x^2 - 10000$. (ii)

Marginal profit $P'(x) = 1500 - 2x$. (iii) For max profit, $P'(x) = 0 \Rightarrow 1500 - 2x = 0 \Rightarrow x = 750$ tanks.

$P''(x) = -2 < 0$, confirms maximum. Maximum profit

$$P(750) = 1500(750) - (750)^2 - 10000 = 1125000 - 562500 - 10000 = ₹5,52,500. \text{ **Q38.** (i) Magnitude of DRs of Plane}$$

$$1 = \sqrt{2^2 + (-1)^2 + 2^2} = \sqrt{9} = 3. \text{ DCs are } \frac{2}{3}, -\frac{1}{3}, \frac{2}{3}. \text{ (ii) Angle}$$

$$\theta = \cos^{-1} \left(\frac{(2)(3) + (-1)(2) + (2)(6)}{\sqrt{4+1+4}\sqrt{9+4+36}} \right) = \cos^{-1} \left(\frac{6-2+12}{3 \times 7} \right) = \cos^{-1} \left(\frac{16}{21} \right). \text{ (iii) Let paths intersect. Path 1 general point:}$$

$$(2\lambda + 1, -\lambda + 2, 2\lambda + 3). \text{ Path 2: } (3\mu + 2, 2\mu + 4, 6\mu + 5). \text{ Equate x and y: } 2\lambda - 3\mu = 1 \text{ and}$$

$$-\lambda - 2\mu = 2 \Rightarrow \lambda = -2\mu - 2. \text{ Substitute into x: } 2(-2\mu - 2) - 3\mu = 1 \Rightarrow -7\mu = 5 \Rightarrow \mu = -5/7. \text{ Then}$$

$$\lambda = -4/7. \text{ Check z coordinates: } 2(-4/7) + 3 = 13/7, \text{ and } 6(-5/7) + 5 = 5/7. \text{ Since } 13/7 \neq 5/7, \text{ they do not intersect.}$$

They are skew lines.